

Slicing and Grabbing: A Natural Interaction Technique for Visualizing Parallel Isolines on 3D Surfaces

Carlos J. Latorre-Rojas* Alexander Rozo-Torres* Javier A. Luzon†‡
Martha Lucía Velasco Morales§ Wilson J. Sarmiento*

* Multimedia Research Group, Universidad Militar Nueva Granada

† Akershus University Hospital, Oslo, Norway

‡ University of Oslo, Oslo, Norway

§ Resonancia Magnética San José, Bogotá, Colombia

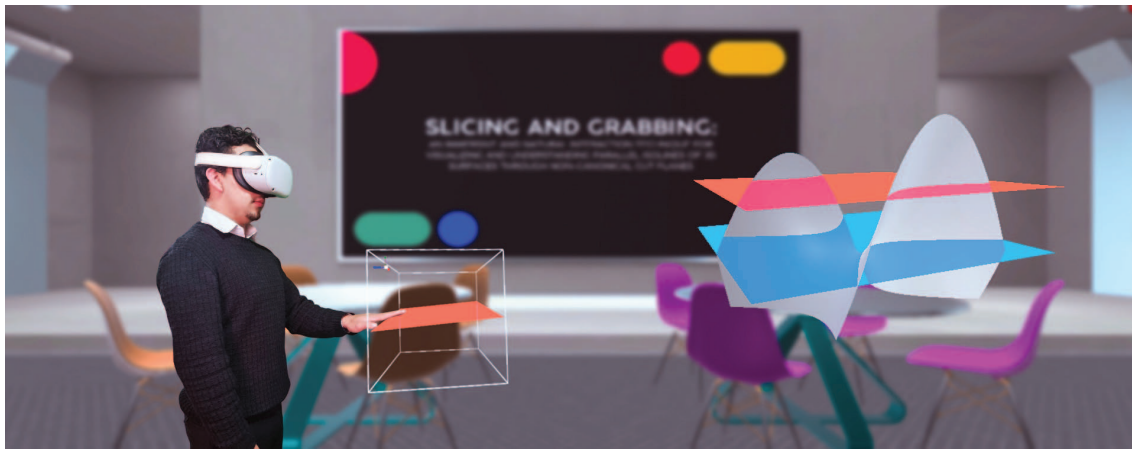


Figure 1: A user is interacting with Slicing and Grabbing 3D interaction flow proposals. It sees in the user front that the widget is provided for cutting plane setup tasks. All interactions are performed using hand gesture interaction.

ABSTRACT

Spatial compression is a crucial skill for a mathematical or engineering student and anyone working with 3D surfaces. Consequently, a well-known issue in math education is that individuals who do not attain a suitable level of spatial compression may encounter difficulties when dealing with more advanced topics. To address this, 3D surface isolines generation is an effective task for honing spatial compression skills. Immersive visualization presents potential alternatives to facilitate the acquisition of spatial compression. This work introduces a 3D interaction flow designed to cut, visualize, and comprehend isolines of 3D surfaces. The proposed technique provides gesture interaction to carry out the task of manipulating 3D surfaces, defining cut planes, and manipulating slices. This proposal offers the option of defining non-canonical project planes to take advantage of the benefits of immersive visualization.

Index Terms: Human-centered computing—Interaction paradigms—Mixed/augmented reality; Human-centered computing—Interaction paradigms;

1 INTRODUCTION

Understanding 3D surfaces and their spatial relationships with themselves or other 3D structures is fundamental to spatial comprehension skills. Some tasks favor that a student acquires these skills. It is the case to isolines marking-out and visualization, which are cuts of a 3D surface by a set of parallel planes. Another way to define

the isolines of a surface is a set of projections, each over a parallel plane. Isolines are widely used for interpreting 3D information because they provide suitable spatial insights into a surface. However, understanding how to generate a set of isolines can be challenging to obtain, requiring an understanding of the physical space by plane projections, which can be challenging for students or trainees of different areas, such as civil, mechanical, and electrical engineering or medical imaging visualization and finite element analysis. While isolines are widely used in different disciplines, those may be complex to interpret for trainees and people not related to the engineering fields [2]. An example of an application of isolines is terrain contour lines that represent a 3D surface in 2D representation. A generalization is the isosurfaces, which are parallel surfaces generated by the protection of high dimensional data when marching cubes is a well-known algorithm for their generation.

Nowadays, computer graphics tools for plotting and interactive visualization represent a valuable resource for contributing to obtaining spatial comprehension skills. For example, visualization techniques such as direct surface rendering with isolines projection facilitate understanding three-dimensional information. However, desktop alternatives have relevant limitations to reach some understanding goals. Thus, identifying proportions and internal features still results in challenges, just as to finding insights [3].

This work proposes a 3D interaction flow called Slicing and Grabbing, allowing a user to generate, visualize, and understand parallel isolines of 3D surfaces through natural hand gestures and immersive visualization. Slicing and Grabbing integrates traditional 3D interaction techniques in a compressive workflow that allows users to interact with a 3D surface and slice it with cut planes so that isoline generation concepts arise naturally while users enjoy the

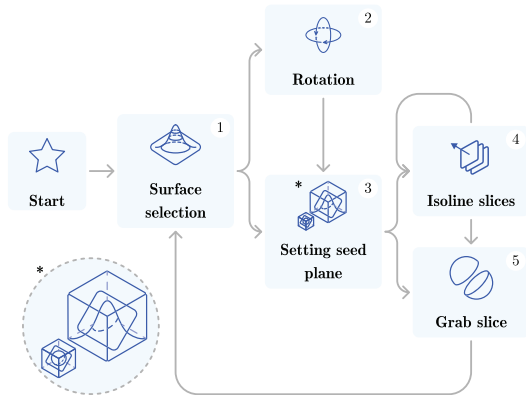


Figure 2: 3D interaction flow of the Slicing and Grabbing.

immersive experience.

2 INTERACTION PROPOSAL

Understanding and comprehending an abstract concept is a cognitive path that can be achieved differently. For this reason, our proposal, rather than a specific interaction technique, is an interactive flow, a path directed toward understanding contour line generation from a 3D surface. The 3D interaction flow proposed is composed of five (5) steps sketched in Figure 2.

In the first step (2), the user chooses the surface target. The options (Surface Selection) are presented in order of complexity, i.e., sphere, ellipsoid, elliptic paraboloid, and hyperbolic cylinder hyperbolic paraboloid. This proposal originates in a real task, interacting with anatomical structures to surgical planning medical tasks [1]. For these reasons, a real heart surface was included to illustrate the real application of the proposal. In the second step (2), the user interacts with the surface chosen and rotates it freely. This step aims for a user to become familiar with the surface. The next steps represents the core *Slicing and Grabbing* proposal: setting the Seed Plane (3) to set up plane orientation and position. Subsequently, they generate the remainder Isoline Slices (4) parallel to the seed plane. In this, a way non-canonical planes set is define to generate the isolines. Finally, the Grab Slice (5) confirms how the 3D model should be sliced and allow the grabbing task for understanding the spacial information.

Most tasks of interaction with the surface are performed using a voodoo-doll technique [5]. Thus, a 3D widget is displayed in the user front to facilitate the surface interaction. In the rotation task, a sphere widget is provided to the user to grab and rotate it. Similarly, a “plane setup” widget” is provided to the user to facilitate the task of defining the plane seed and cutting the plane set. The Figure 1 shows the “plane setup” widget”. Only the Grabbing task is done directly with the surface; the user takes a selected slice using the virtual hand technique.

All Slicing and Grabbing interaction flow tasks are performed with natural hand gesture interaction. Figure 3 shows the gesture set used. The Thumb-up gesture is used for any confirmation task. The swipe gesture is performed with the hand open vertically, moving to the right or left. This gesture is used to navigate in a chosen task. The plane setup task is executed with an open hand gesture in a horizontal position. The user rotates his/her hand to define the plane orientation and moves the hand up and down to place the plane position. To grab a slice, the user does the Pinch gesture.

Slicing and Grabbing were implemented for the *Oculus Quest 2* system over the standard *OpenXR standard*. The gesture recognition algorithms were written using a hand-tracking engine provided by

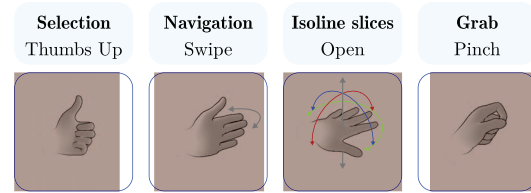


Figure 3: Natural hand gesture set using in Slicing and Grabbing.

Oculus SDK. Unity was the development environment, and VTK provided visualization features through open source plugin [4].

3 DISCUSSION AND FINAL REMARKS

This work integrates traditional 3D interaction techniques to propose an interaction concept supported by gesture interactions, allowing users to generate a set of isolines on a 3D surface. Then, Slicing and Grabbing enable the user to 1) choose a target 3D surface, 2) visualize and rotate the surface, 3) define a seed plane for the isolines, 4) generate parallel cut planes, and 5) grab to examine a slice of the surface. Thus, the interaction flow allows the student to understand the concept of parallel isolines and surface projection, improving his/her spatial comprehension skills. Preliminary tests with students of our lab show that slicing and grabbing are highly attractive for visualizing 3D information, and the gesture interactions are intuitive and fun. This technique is being refined for use in a surgical planning context with the support of two physicians with more than 50 and 30 years of experience in surgery and radiology, respectively. While slicing 3D models is a classical manipulation task, adding natural interaction represents an intuitive method for analyzing datasets such as diagnostic images, facilitating appreciation of the representation of internal anatomical structures. This promising application requires further evaluation for preoperative surgical planning.

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REFERENCES

- [1] C. J. Latorre-Rojas, A. Rozo-Torres, and W. J. Sarmiento. Interaction models for surgical planning in extended reality. challenges in radiologist-surgeon communication. In *2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 143–146, 2023. doi: 10.1109/VRW58643.2023.00036 2
- [2] K. Słomska-Przech and I. M. Golebiowska. Do different map types support map reading equally? comparing choropleth, graduated symbols, and isoline maps for map use tasks. *ISPRS International Journal of Geo-Information*, 10(2):69, 2021. 1
- [3] S. A. Sorby. Educational research in developing 3-d spatial skills for engineering students. *International Journal of Science Education*, 31(3):459–480, 2009. doi: 10.1080/09500690802595839 1
- [4] G. Wheeler, S. Deng, N. Toussaint, K. Pushparajah, J. A. Schnabel, J. M. Simpson, and A. Gomez. Virtual interaction and visualisation of 3d medical imaging data with vtk and unity. *Healthcare technology letters*, 5(5):148–153, 2018. doi: 10.1049/htl.2018.5064 2
- [5] P. Yao, V. Lympouridis, and M. Zyda. Virtual equipment system: Face mask and voodoo doll for user privacy and self-expression options in virtual reality. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 747–748, 2021. doi: 10.1109/VRW52623.2021.00256 2